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Introduction to Signal Intelligence



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NEED

What is SIGINT, and why is it important in military operations?

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Hours *with break and quiz*

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ROADMAP



EXPLAIN

- General Objectives
- Specific Objectives
- Definition of Intelligence
- Definition of SIGINT & Categories of SIGINT
- Direction Finding
- SIGINT Capability of Army Units
- Advantages of SIGINT
- Disadvantages of SIGINT
- SIGINT Cycle
- Function

APPLY thru check on learning

SUMMARIZE let the students summarize the lesson

ASSESS thru written examination

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OUTCOME

The student should be able to EXPLAIN and DESCRIBE SIGINT operations.

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CLASSROOM RULES









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SAFETY PRECAUTIONS







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SAFETY PRECAUTIONS

What should I do? In case of an earthquake:

Earthquakes strike without warning



REMAIN CALM
Remain calm and try to calm others. Be aware of your surroundings.

WATCH OUT
Watch for falling plaster, light fixtures, ceiling tiles and other objects.

STAY AWAY
Stay away from windows, mirrors and heavy objects. Do not run outdoors.

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REFERENCE

PAM 2-00 PA Intel Manual

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General Objective

The whole PA SIGINT program is anchored on the principle of providing real time intelligence to the infantry Brigades and Battalions in support of the AFP missions.

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Specific Objectives

- To maintain the effective Army-Wide Radio HF/VHF/UHF intercept operation and radio direction finding (RDF) capability.
- To continuously upgrade the SIGINT Compartments responsive to the tactical requirements of the operating units.
- To develop strategic monitoring site in line with AFP external defense posture.
- To set-up Army-Wide secured intelligence communication network.
- To sustain efficiency through professional and technical development/training of PA/AFP SIGINT personnel.
- To equip the students with the necessary knowledge, skills, and attributes in SIGINT Operations

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INTELLIGENCE - is the product resulting from the collection, evaluation, analysis, integration of all available information which concerns one or more aspects of foreign nations or of area of operations and which is immediately or potentially significant to military planning and operations.

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WHAT IS SIGINT?

..it means Signals Intelligence.

..is intelligence gathering by interception of signals, whether communications (COMINT) between people or from electronic (ELINT) signals not directly used in communication.

..as sensitive information is often encrypted, SIGINT in turn involves the use of **cryptanalysis** to decipher messages and **traffic analysis** to re-integrate information.

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2 CATEGORIES OF SIGINT

2. Electronic Intelligence (ELINT)

- intelligence gathering by use of electronic sensors and various electronic surveillance systems.

Capable of doing some of the following:

- ✓ *Direction finding to a target in signals intercept*
- ✓ *Jamming*
- ✓ *Communication intercept through traffic analysis of electronic emissions*

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2 CATEGORIES OF SIGINT

1. Communication Intelligence (COMINT)

- engages in dealing with messages or voice information derived from the interception of foreign communication.

Capable of revealing some of the following:

- ✓ *Who is transmitting?*
- ✓ *Location*
- ✓ *Text interception*
- ✓ *Voice interception*

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SIGINT CAPABILITY OF ARMY UNITS

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RADIO DIRECTION FINDER SYSTEM (RDF)



Radio Receiver and Bearing Processor



Doppler Antenna

- It is used for radio communication transmission counter measures to monitor enemy activities and locate their estimated locations.

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Advantages &Disadvantages of SIGINT

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ADVANTAGES OF SIGINT

- **A**ccurate
- **T**imely
- **R**elevant

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DISADVANTAGES OF SIGINT

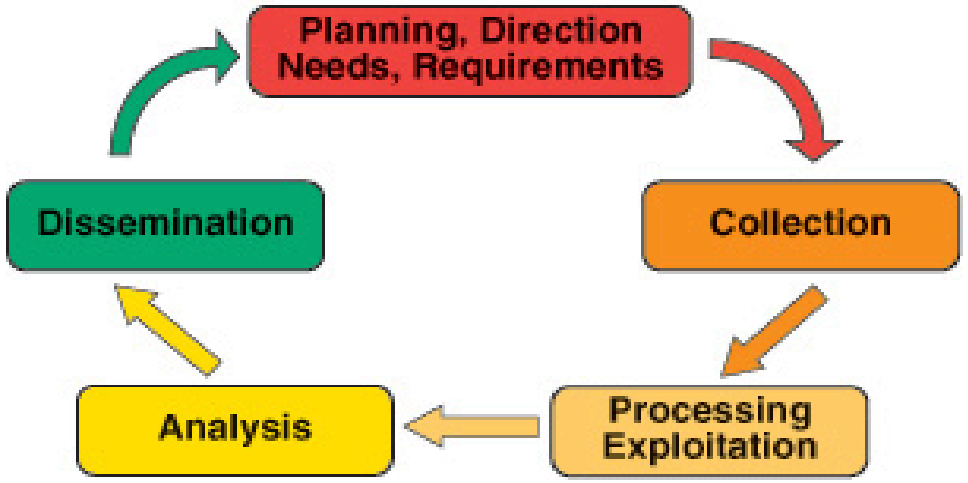
- SIGINT sometimes cannot stand alone without substantiation from other intelligence disciplines.
- SIGINT cannot be clarified from the source.
- If security is compromised, the enemy can counter the government efforts with disinformation or deception.

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SIGINT CYCLE

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Intelligence Cycle

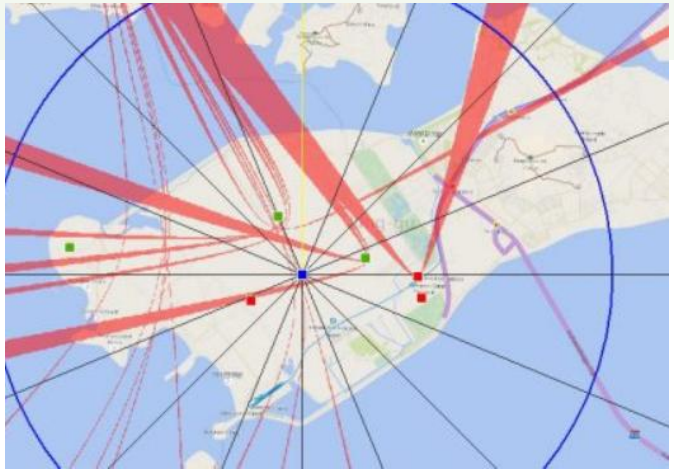


- ✓ **Requirements** – the SIGINT operators must understand the mission and the **direction needs** in order to **plan** and apply appropriate collection mechanisms.
- ✓ **Collection** – proper procedure of gathering information
- ✓ **Processing/Exploitation** - intercepted communication traffics and estimated direction of the enemy
- ✓ **Analysis** – converted collected SIGINT data into intelligence
- ✓ **Dissemination** – Timely production of intelligence to those who need to know only

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FUNCTION

- Conduct of direction finding (DF) operation and provide technical control analysis of SIGINT operation.
- Conduct advanced analysis on intercepted communications and non-communications signals information.





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✓ CHECK ON LEARNING

is intelligence gathering by interception of signals, whether communications (COMINT) between people or from electronic (ELINT) signals not directly used in communication.

SIGINT

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✓ CHECK ON LEARNING

2 CATEGORIES OF SIGINT

Communication Intelligence (COMINT)
Electronic Intelligence (ELINT)

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SUMMARY

1. General Objectives
2. Specific Objectives
3. Definition of Intelligence
4. Definition of SIGINT & Categories of SIGINT
5. Direction Finding
6. SIGINT Capability of Army Units
7. Advantages of SIGINT
8. Disadvantages of SIGINT
9. SIGINT Cycle
10. Function

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ASSESSMENT

Written Examination (10 items – identification)

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OUTCOME

By this time students were able to EXPLAIN and DESCRIBE SIGINT operations.

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FEEDBACK

Discuss the result of the assessment

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FUTURE

Next session is about Introduction to HUMINT

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THANK YOU
AND
GOOD DAY